

CLOTTING CASCADE

by

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OBJECTIVES OF LECTURE

At the end of lecture students should be able to

- Describe the steps of Hemostasis
- Discuss the functions of platelets
- Explain the process of platelet plug formation
- Discuss the Intrinsic and Extrinsic pathway of clotting
- Enlist the Clotting factors
- Discuss the process of fibrinolysis
- Discuss the factors which prevent intravascular coagulation

Steps of Hemostasis

1. vasoconstriction
2. primary hemostasis(platelet plug formation)
3. Secondary hemostasis(clot formation)
4. Clot retraction
5. Fibrinolysis

1. Vasoconstriction

1. Local myogenic spasm

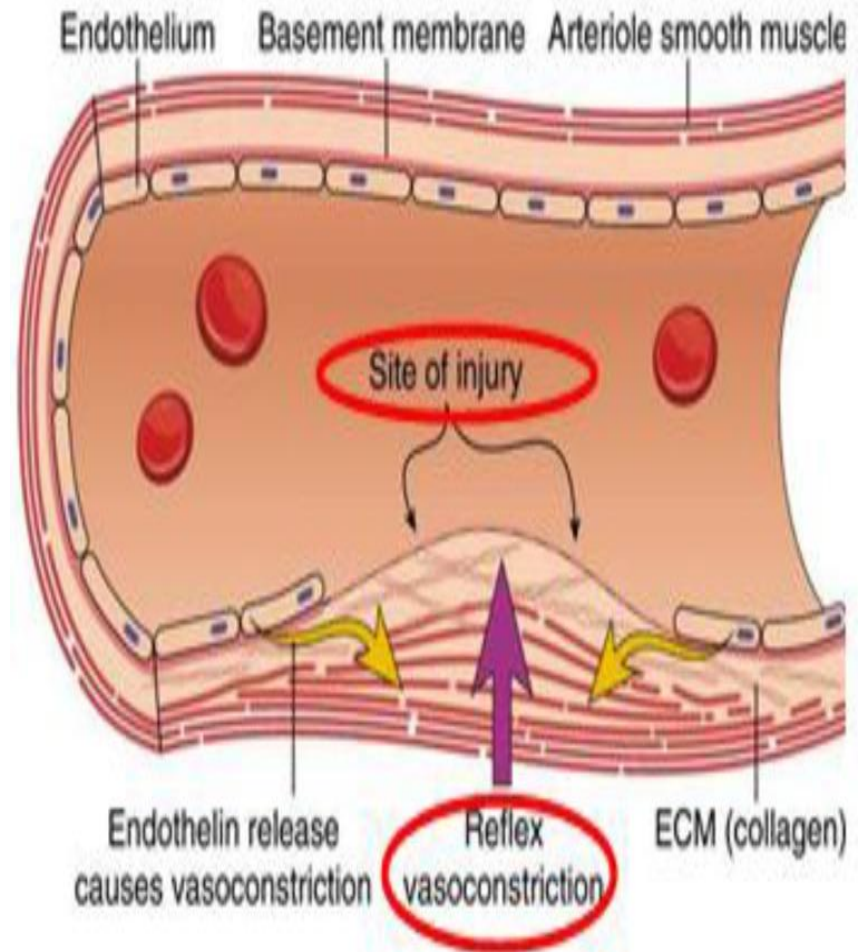
- when direct injury to smooth muscles of vessel

2. Local factors

- for example Endothelin which causes contraction of smooth muscles

3. nervous reflexes

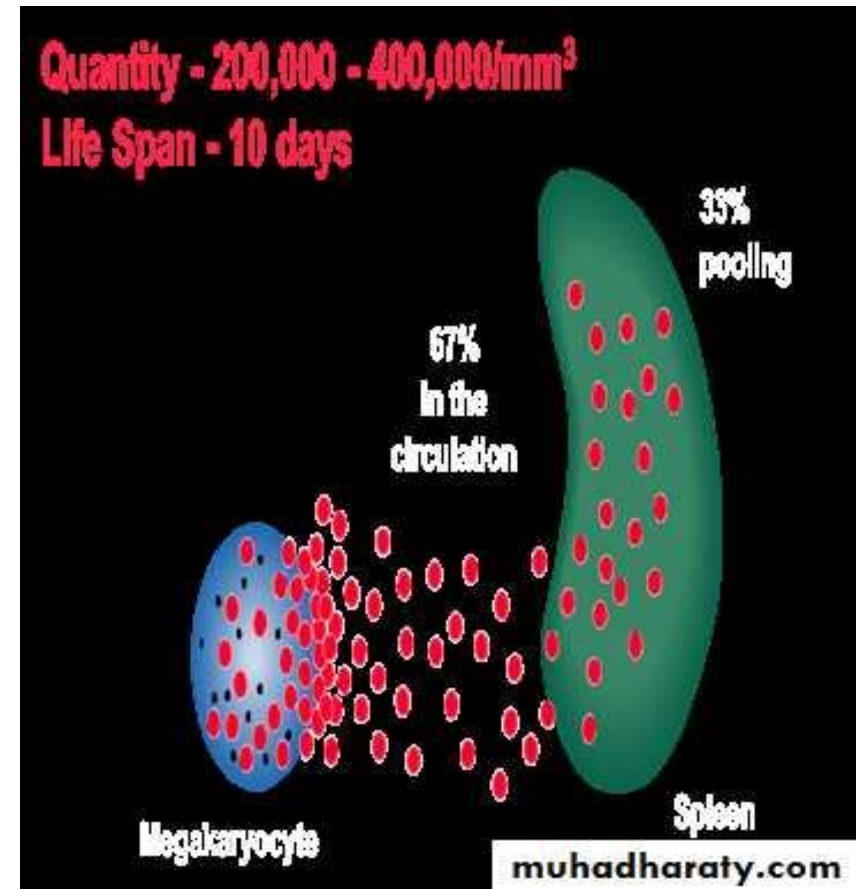
- from pain receptors



2.Primary Hemostasis

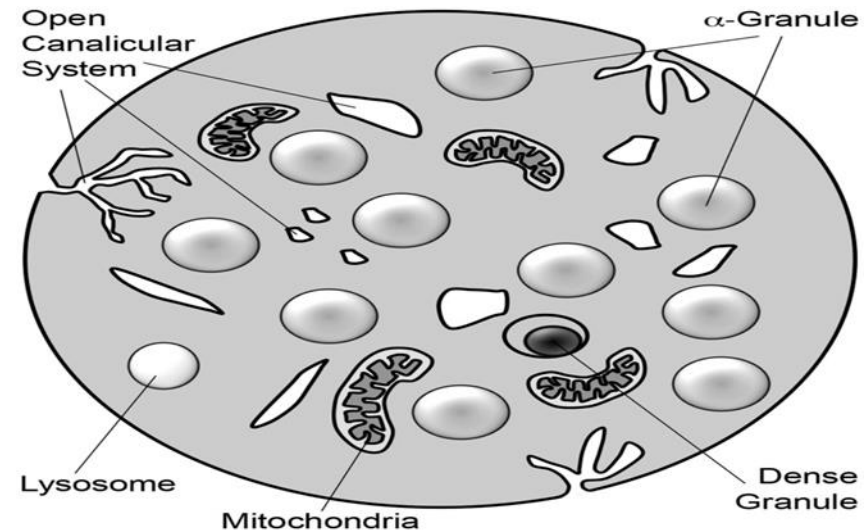
(Formation of the platelet plug)

- Fragmentation of Megakaryocytes, in bone marrow, form platelets by pinching off bits of cytoplasm.
- 60%and 70% are in circulation
- Remainder in spleen.

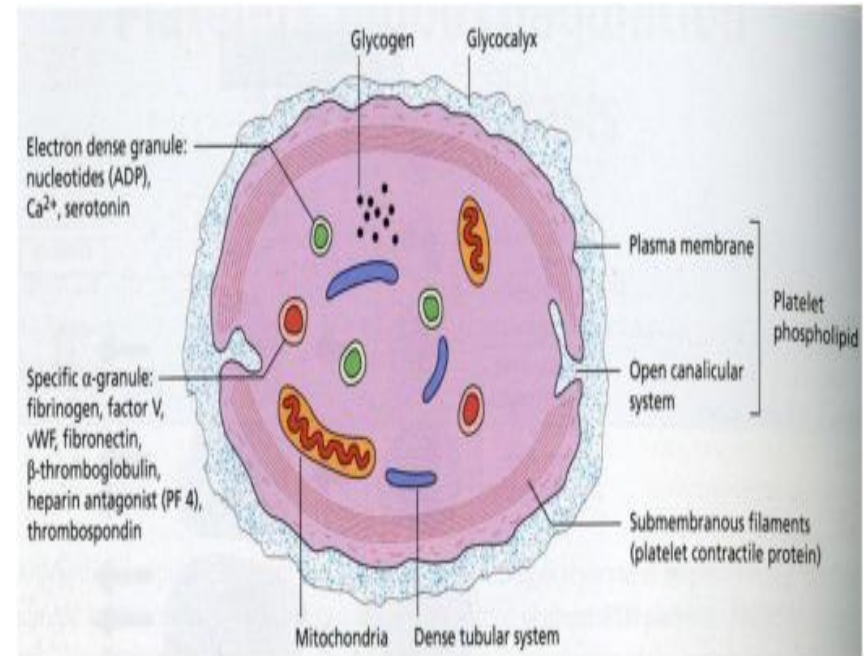


STRUCTURE OF PLATELETS

- Platelets have a microtubules around their periphery with canalicular system.
- Cytoplasm contains actin, myosin, thrombosthenin
- residuals of endoplasmic reticulum and the Golgi apparatus that synthesize various enzymes and store large quantities of calcium ions
- mitochondria and enzyme systems forming adenosine triphosphate (ATP) and adenosine diphosphate (ADP)
- enzyme systems synthesize prostaglandins,
- fibrin-stabilizing factor stabilizes clot
- a growth factor that causes vascular endothelial cells, vascular smooth muscle cells, and fibroblasts to helps repair damaged vascular walls

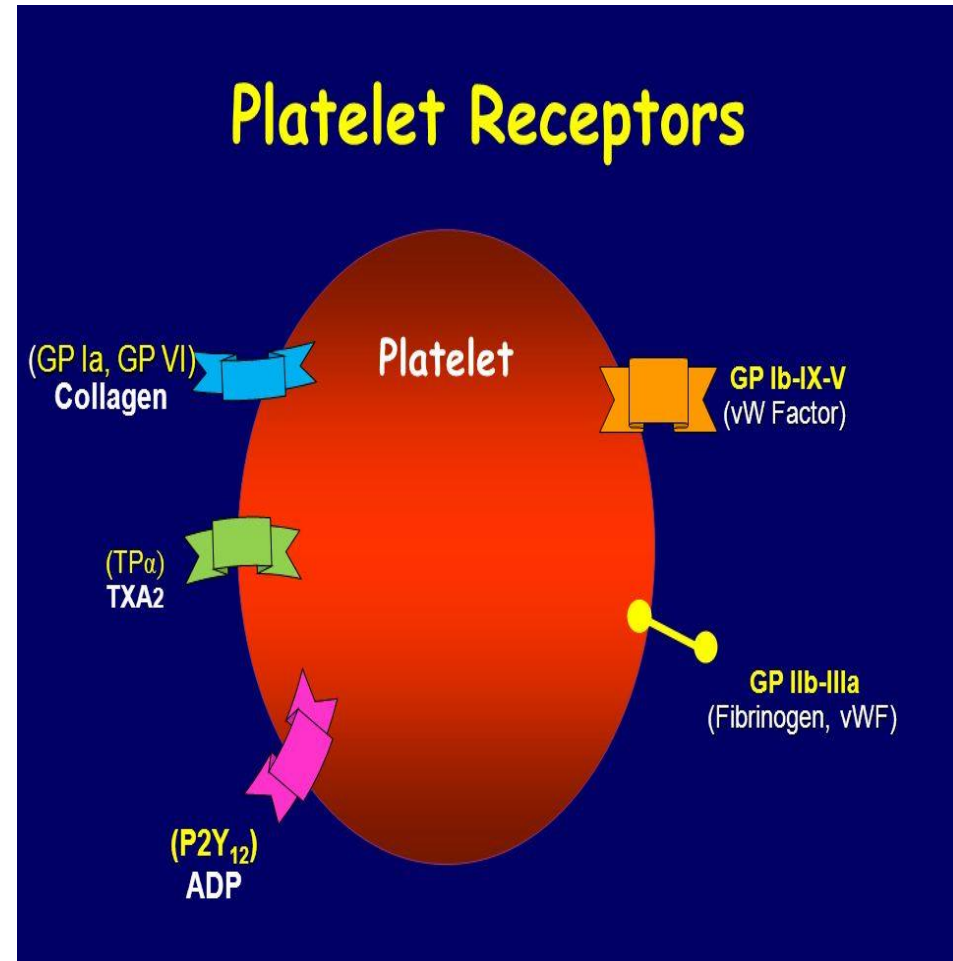


- On the platelet cell membrane surface is a coat of glycoproteins that repulses adherence to normal endothelium
- the platelet membrane contains large amounts of phospholipids that activate multiple stages in the blood-clotting process,
- half-life in the blood of 8 to 12 days
- eliminated from the circulation mainly by the tissue macrophage system..



ultrastructure of platelet

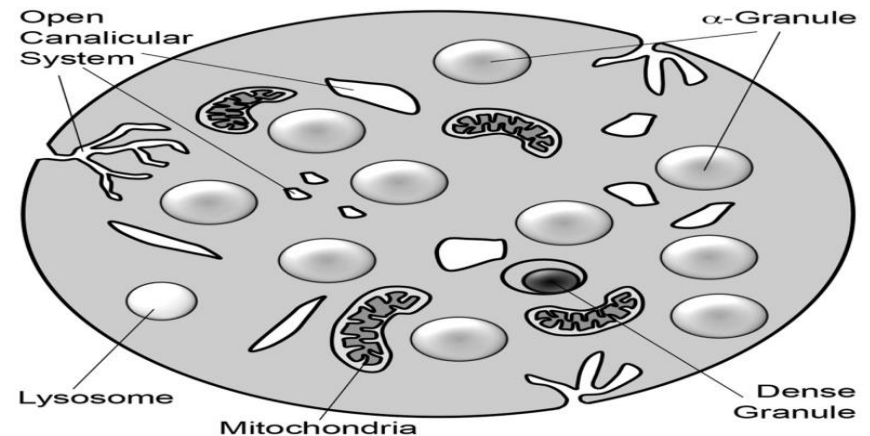
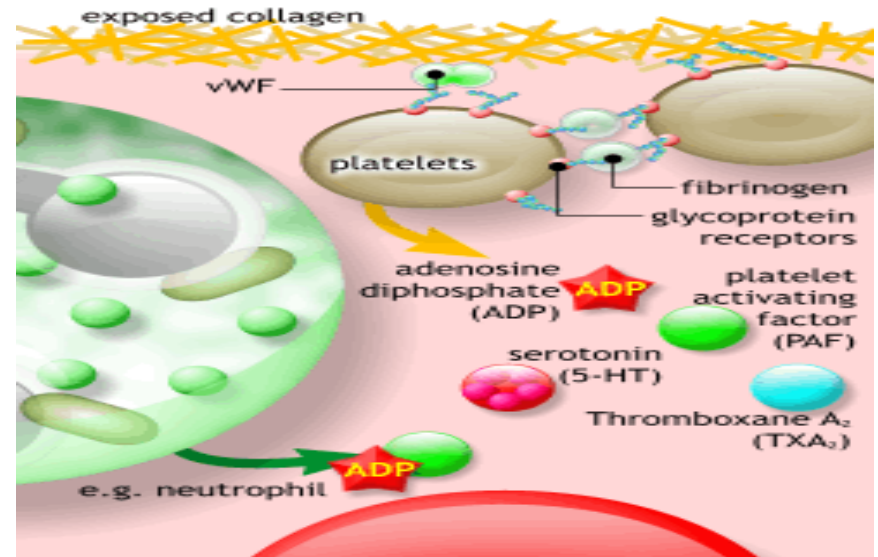
- Membranes contain receptors for collagen, ADP, von Willebrand factor and fibrinogen.



Two types of granules:

Dense granules: which contain the non protein substance e.g. ADP,ATP,calcium and serotonin .

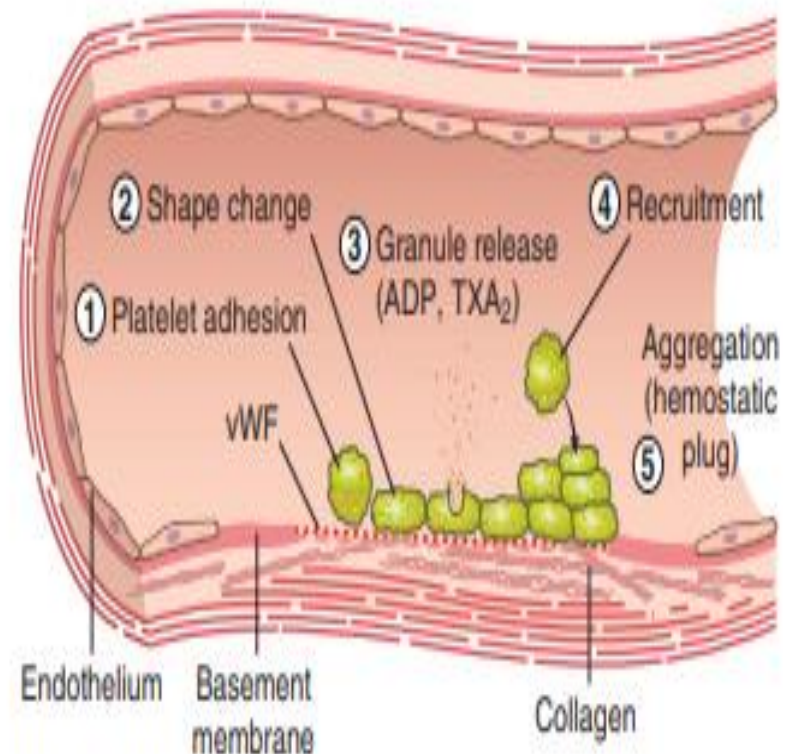
Alpha granules: containing secreted proteins e.g. PDGF and transforming growth factor- β .



ROLE OF PLATELETS **IN HEMOSTASIS**

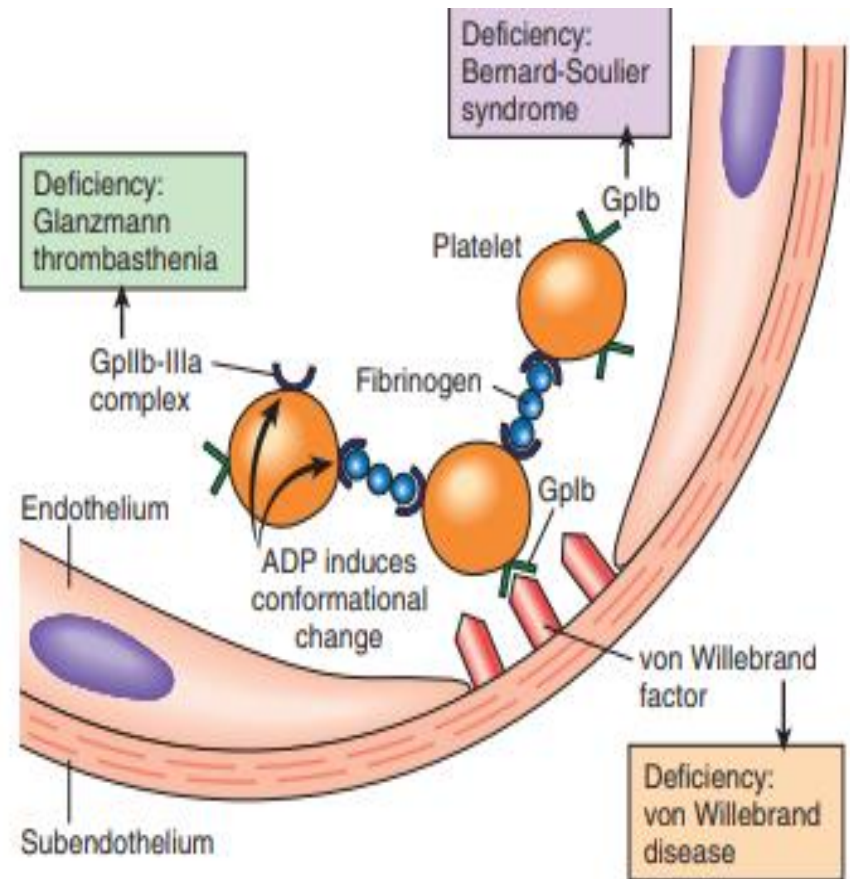
Endothelial injury leads to

- Platelet adhesion.
- Platelet activation and secretion
- Platelet aggregation.

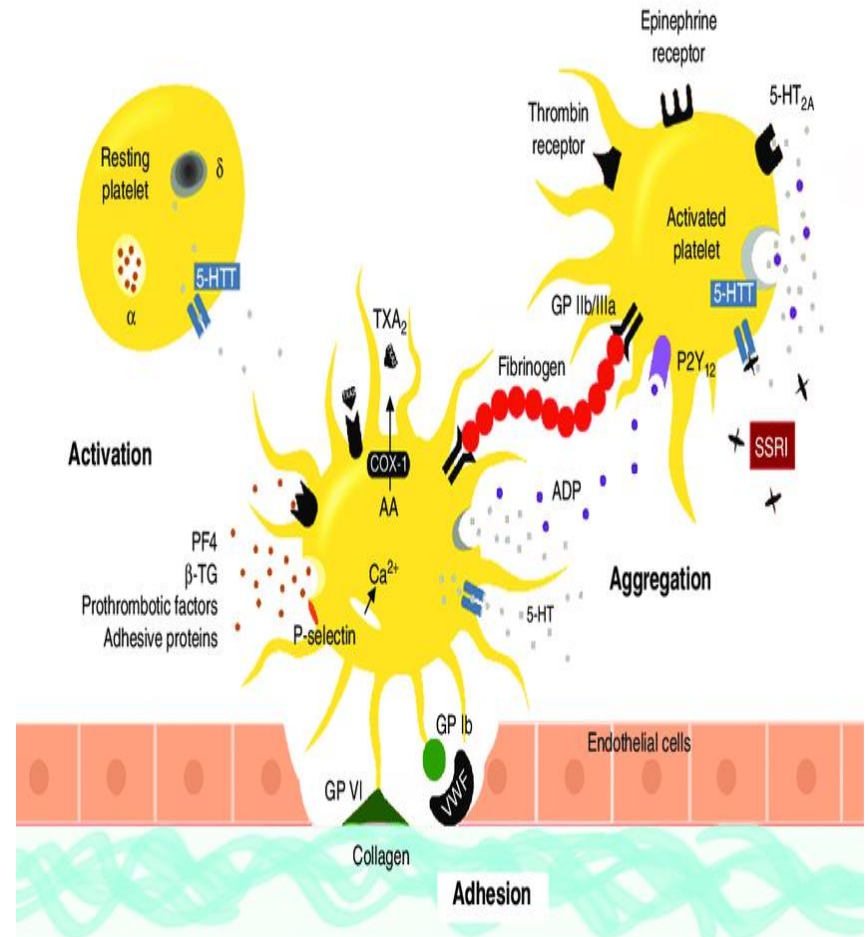


Platelet adhesion

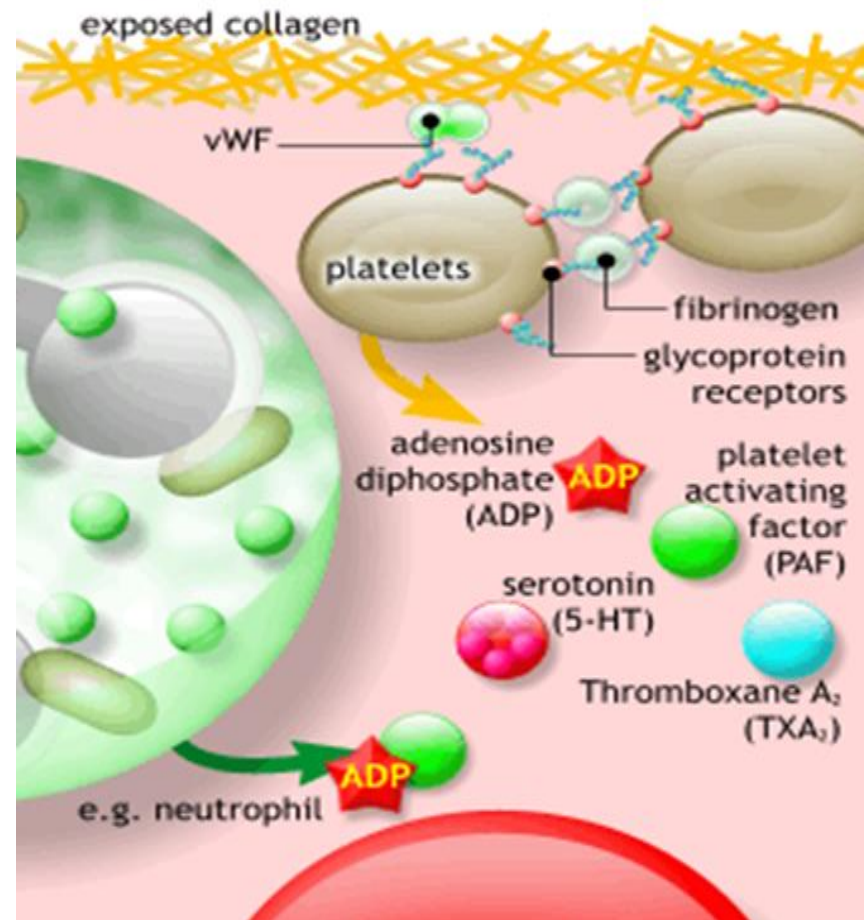
- It is mediated via interactions with vWF, which acts as a bridge between the platelet surface receptor glycoprotein Ib (GpIb) and exposed collagen

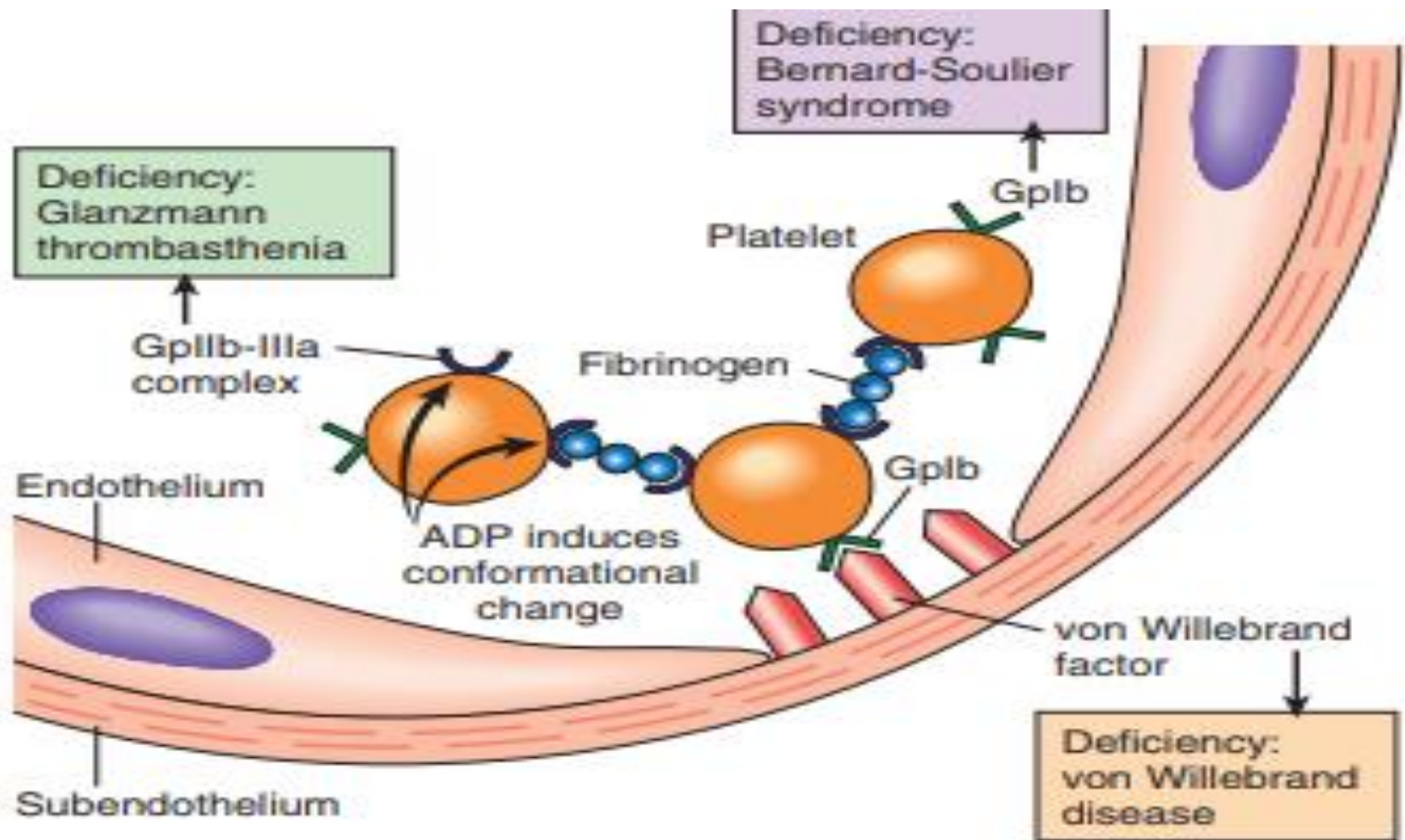


- Platelets rapidly change their shape from disc like to spiky cells after adhesion, which increases surface area and also release the secretion of granules.
- This change is accompanied by alterations in glycoprotein IIb/IIIa that increase its affinity for fibrinogen that forms bridges between adjacent platelets, leading to their aggregation.

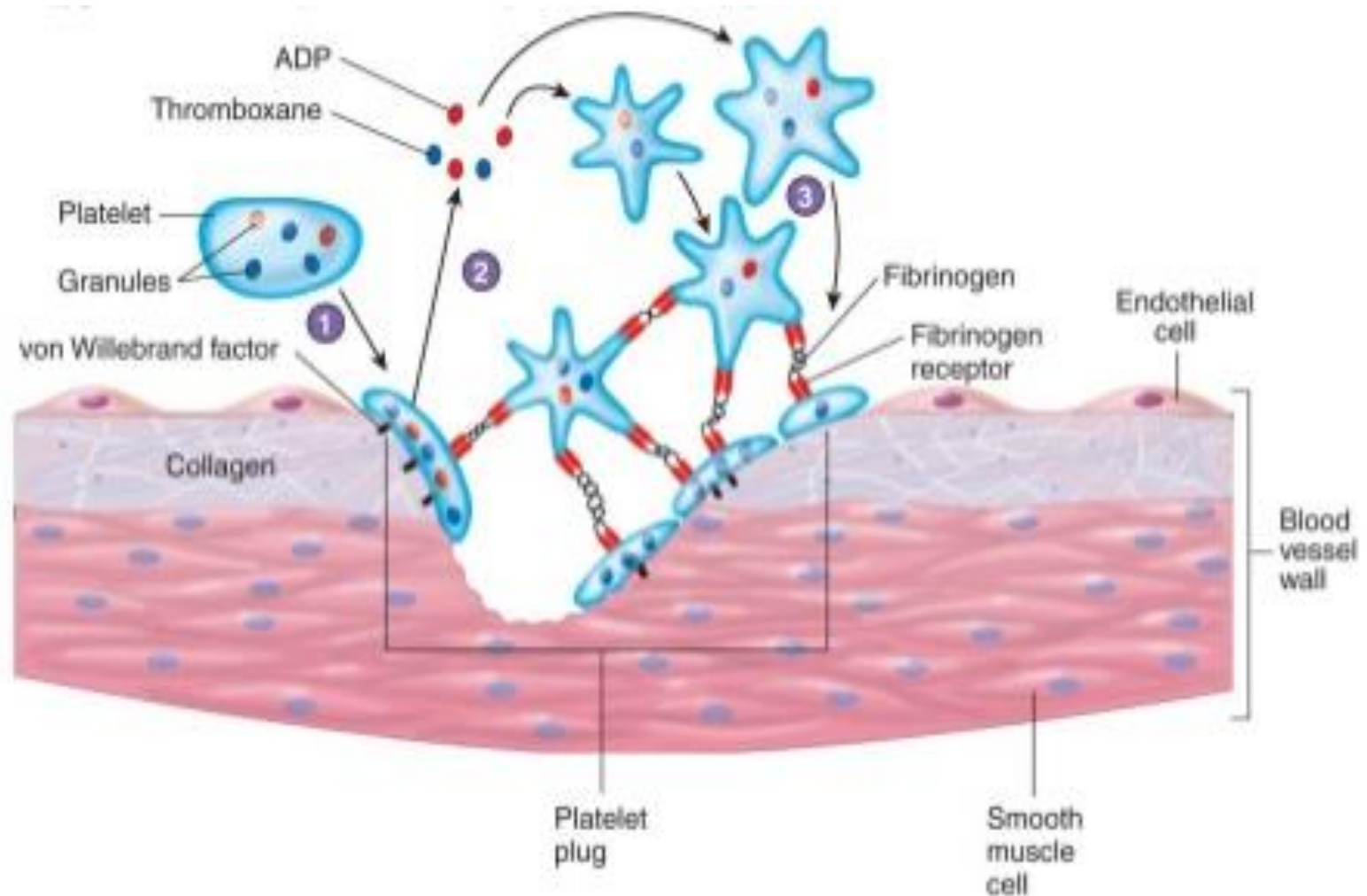


- Activated platelets release adenosine diphosphate (ADP) from their storage granules, which causes the surfaces of nearby circulating platelets to become sticky so that they adhere to the first layer of aggregated platelets and are activated.
- Activated platelets also produce the prostaglandin thromboxane A₂ (TXA₂), a potent inducer of platelet aggregation.
- Note: Aspirin inhibits platelet aggregation by inhibiting cyclooxygenase, a platelet enzyme required for TXA₂ synthesis.





Platelet Plug formation

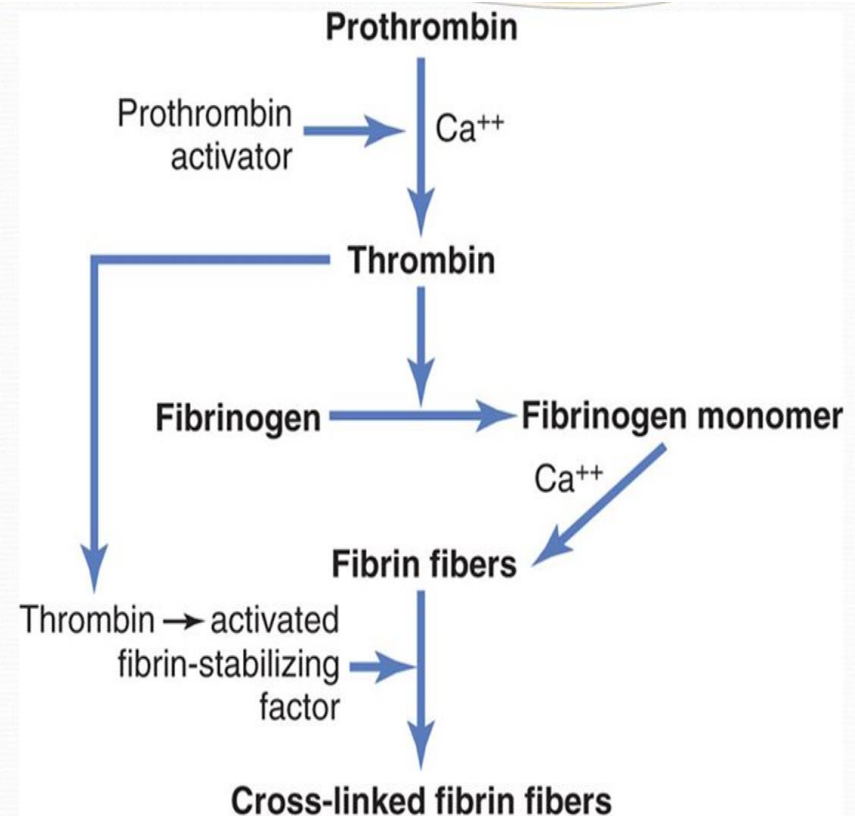


- why does the platelet plug not continue to develop and expand over the surface of the adjacent normal vessel lining?
- A key reason is that ADP released by the activated platelets stimulates the release of prostacyclin and nitric oxide from the adjacent normal endothelium. Both these chemicals inhibit platelet aggregation. Thus, the platelet plug is limited to the defect and does not spread to the nearby undamaged vascular tissue.

3. Secondary hemostasis(clot formation)

Initiation of coagulation: formation of prothrombin activator

- prothrombin activator causes prothrombin conversion to thrombin and all the subsequent clotting steps.
- **Stages of Blood Clotting**
- In general, blood clotting occurs in three stages:
- 1. Formation of prothrombin activator
- 2. Conversion of prothrombin into thrombin
- 3. Conversion of fibrinogen into fibrin.



Clotting Factors:

Clotting Factors	Synonyms
Factor I	Fibrinogen
Factor II	Prothrombin
Factor III	Tissue factor/Tissue thromboplastin
Factor IV	Calcium
Factor V	Proaccelerin/Labile factor/Ac-globulin
<i>Factor VI</i>	
Factor VII	Stable factor/Proconvertin
Factor VIII	Anti-hemophilic factor/Anti-hemophilic factor A/Anti-hemophilic globulin
Factor IX	Christmas factor/Anti-hemophilic factor B/Plasma thromboplastin component
Factor X	Stuart factor/Stuart power factor
Factor XI	Plasma thromboplastin antecedent/Anti-hemophilic factor C
Factor XII	Hageman factor
Factor XIII	Fibrin stabilizing factor

Prothrombin activator is formed by two mechanism

Extrinsic Clotting Mechanism

- chemical outside of blood triggers blood coagulation
- triggered by tissue thromboplastin (not found in blood), formed from the injured tissues
- triggered when blood contacts damaged tissue

Intrinsic Clotting Mechanism

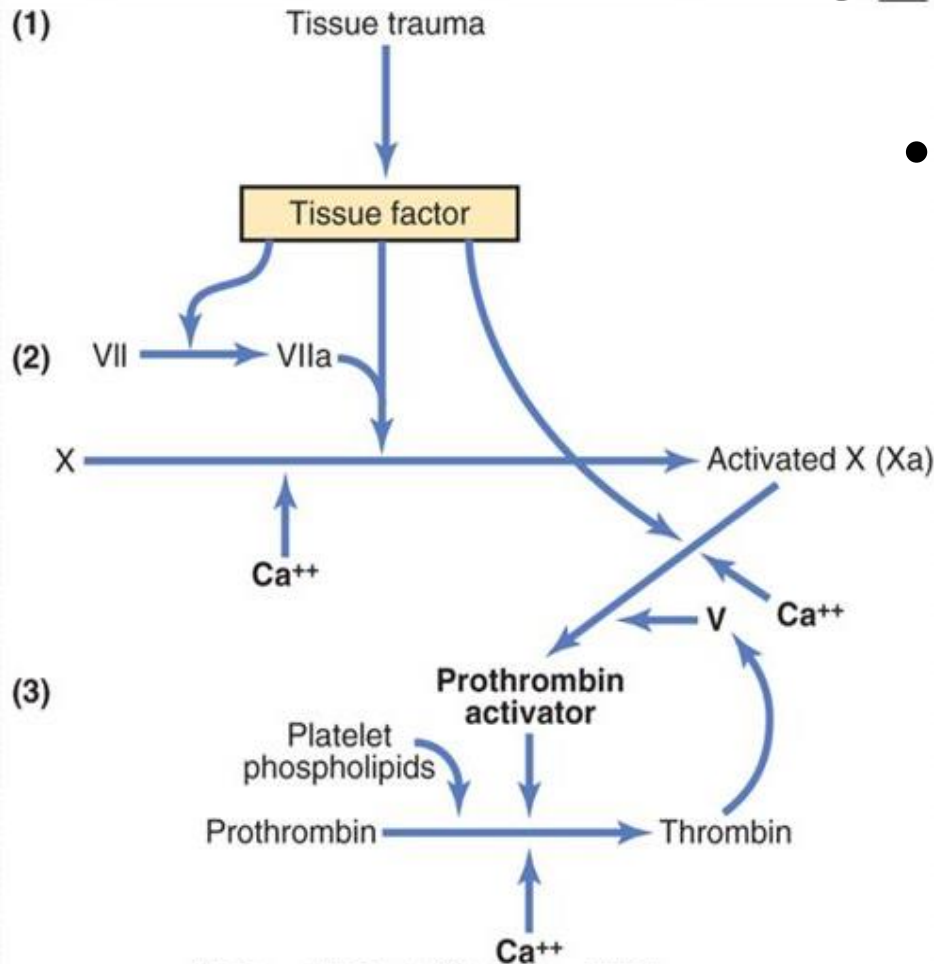
- chemical inside blood triggers blood coagulation
- triggered by Hageman factor (found inside blood)
- triggered when blood contacts a foreign surface

Sequence of Events in Extrinsic Pathway

- i. Tissues that are damaged during injury release tissue thromboplastin (factor III). Thromboplastin contains proteins, phospholipid and glycoprotein, which act as proteolytic enzymes
- .
- ii. Glycoprotein and phospholipid components of thromboplastin convert factor X into activated factor X, in the presence of factor VII.
- iii. Activated factor X reacts with factor V and phospholipid component of tissue thromboplastin to form prothrombin activator. This reaction requires the presence of calcium ions.

Extrinsic clotting mechanism

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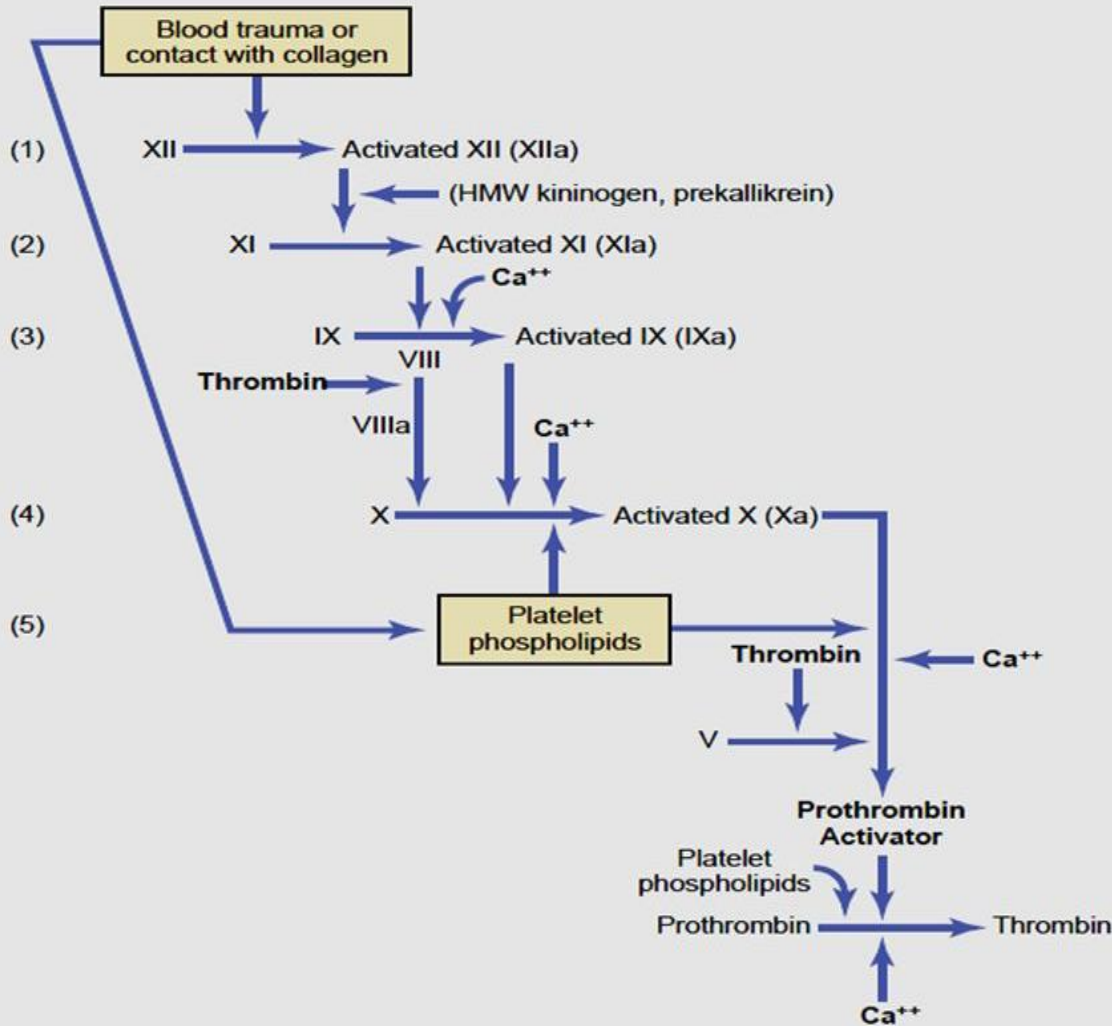


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Sequence of Events in Intrinsic pathway

- **i.** During the injury, the blood vessel is ruptured. Endothelium is damaged and collagen beneath the endothelium is exposed.
- **ii.** When factor XII (Hageman factor) comes in contact with collagen, it is converted into activated factor XII in the presence of kallikrein and high molecular weight (HMW) kinogen.
- **iii.** The activated factor XII converts factor XI into activated factor XI in the presence of HMW kinogen.
- **iv.** The activated factor XI activates factor IX in the presence of factor IV (calcium).
- **v.** Activated factor IX activates factor X in the presence of factor VIII and calcium.
- **vi.** When platelet comes in contact with collagen of damaged blood vessel, it gets activated and releases phospholipids.
- **vii.** Now the activated factor X reacts with platelet phospholipid and factor V to form prothrombin activator. This needs the presence of calcium ions.
- **viii.** Factor V is also activated by positive feedback effect of thrombin .

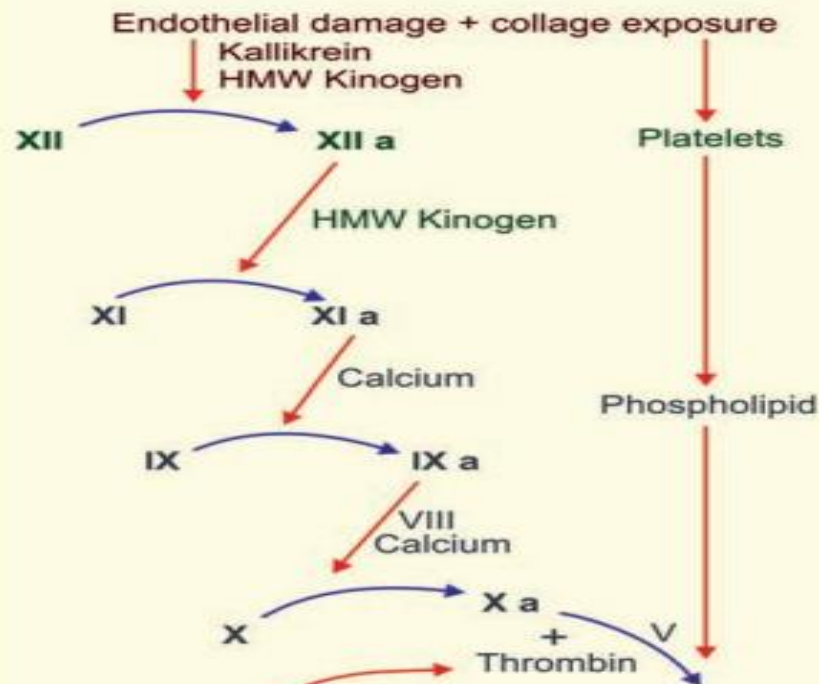
Intrinsic clotting mechanism



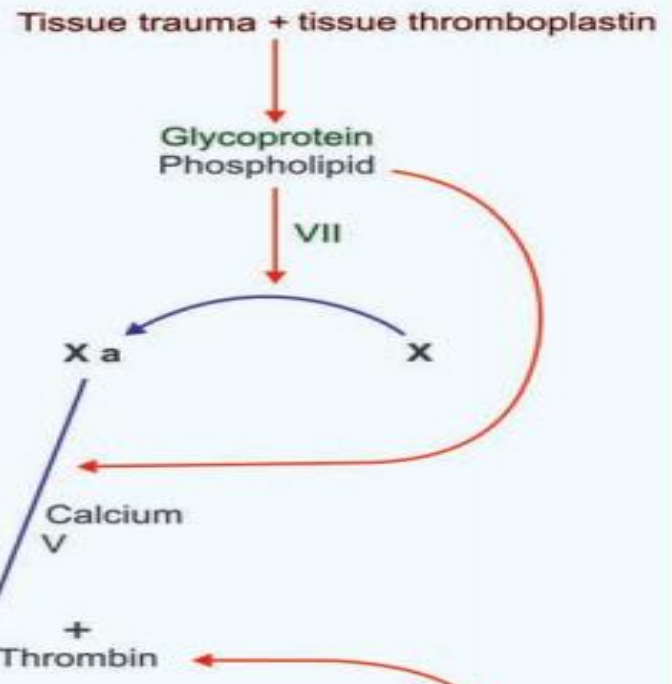
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Stage 1

Intrinsic pathway



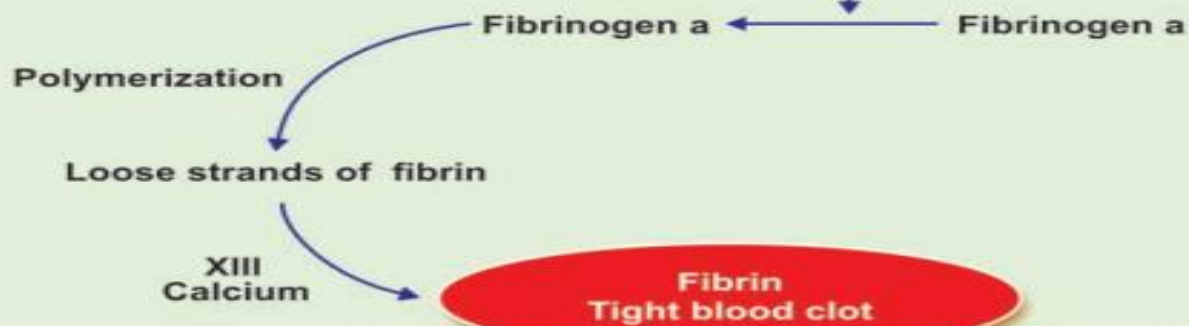
Extrinsic pathway



Stage 2



Stage 3



Blood Coagulation

- **Role of Calcium in Extrinsic and Intrinsic Pathways**
 - a. Except for the first two steps in the intrinsic pathway, calcium ions are required for promotion or acceleration of all blood clotting reactions
 - b. Normally, calcium levels do not fall low enough to affect blood clotting mechanisms

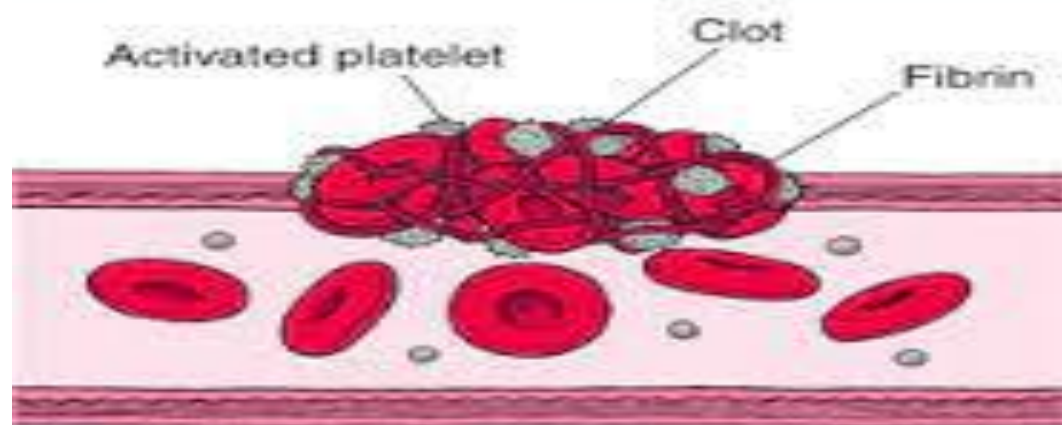
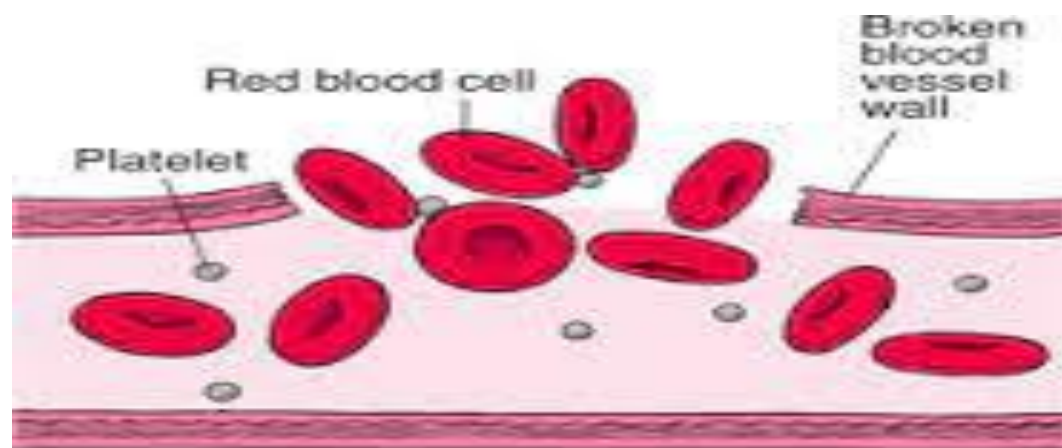
BLOOD COAGULATION

TABLE 14.8 Blood Coagulation

Steps	Extrinsic Clotting Mechanism	Intrinsic Clotting Mechanism
Trigger	Damage to vessel or tissue	Blood contacts foreign surface
Initiation	Tissue thromboplastin	Hageman factor
Series of reactions involving several clotting factors and calcium ions (Ca^{+2}) lead to the production of:	Prothrombin activator	Prothrombin activator
Prothrombin activator and calcium ions cause the conversion of:	Prothrombin to thrombin	Prothrombin to thrombin
Thrombin causes fragmentation, then joining of:	Fibrinogen to fibrin	Fibrinogen to fibrin

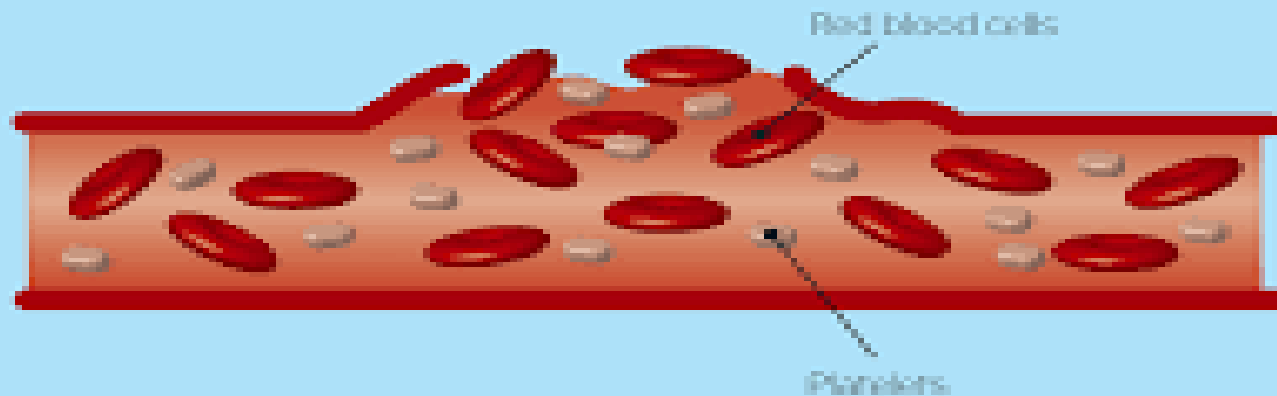
Clot Retraction

- platelets contribute directly to clot contraction by activating platelet thrombosthenin, actin, and myosin molecules, which are all contractile proteins in the platelets
- This action also helps compress the fibrin meshwork into a smaller mass.
- The contraction is activated and accelerated by thrombin, as well as by calcium ions released from calcium stores of the platelets.

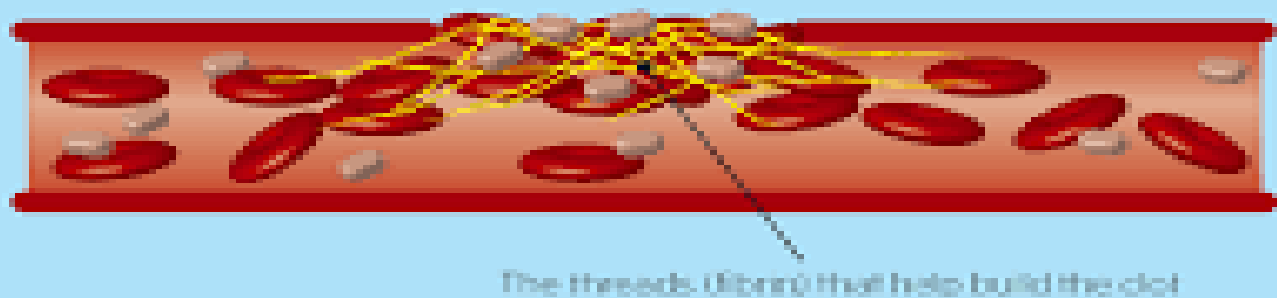


Formation of blood clots

1. Damaged blood vessel wall



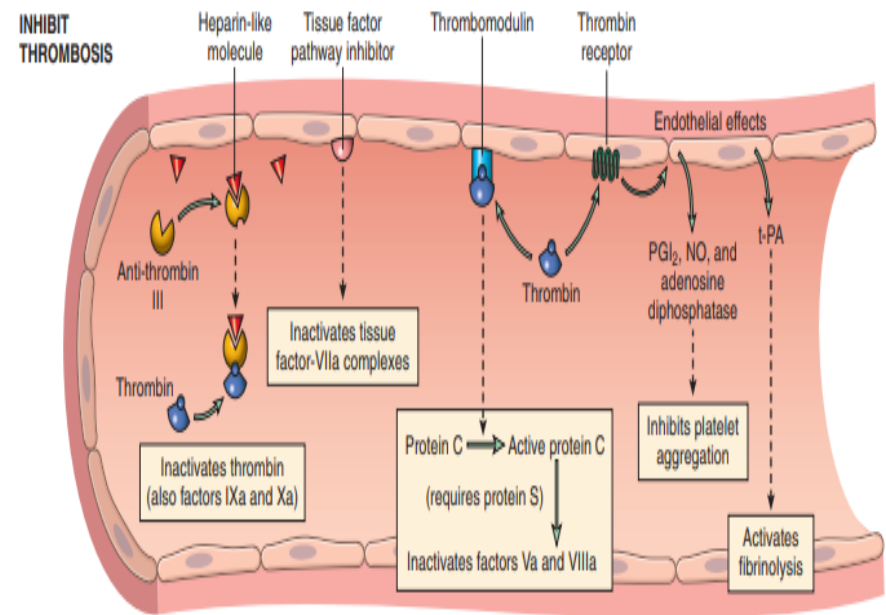
2. Repaired vessel wall



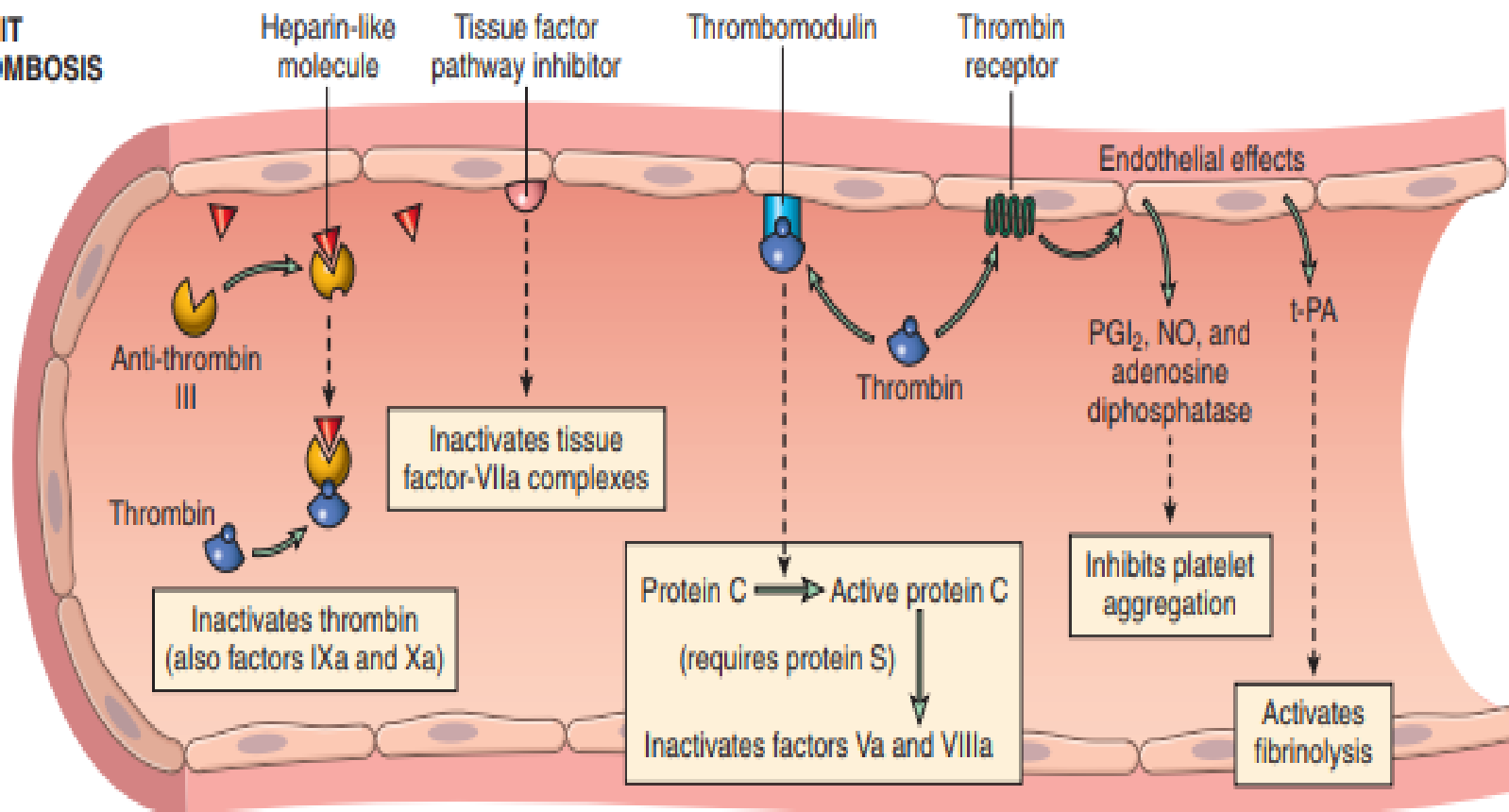
Intravascular Anticoagulants Prevent Blood Clotting in the Normal Vascular System

Endothelial Surface Factors.

- (1) the smoothness of the endothelial cell surface, which prevents contact activation of the intrinsic clotting system
- (2) a layer of glycocalyx on the endothelium repels clotting factors and platelets, thereby preventing activation of clotting
- (3) a protein bound with the endothelial membrane, thrombomodulin, which binds thrombin removing thrombin
- thrombomodulin-thrombin complex also activates a plasma protein, protein C, that acts as an anticoagulant by inactivating activated Factors V and VIII.

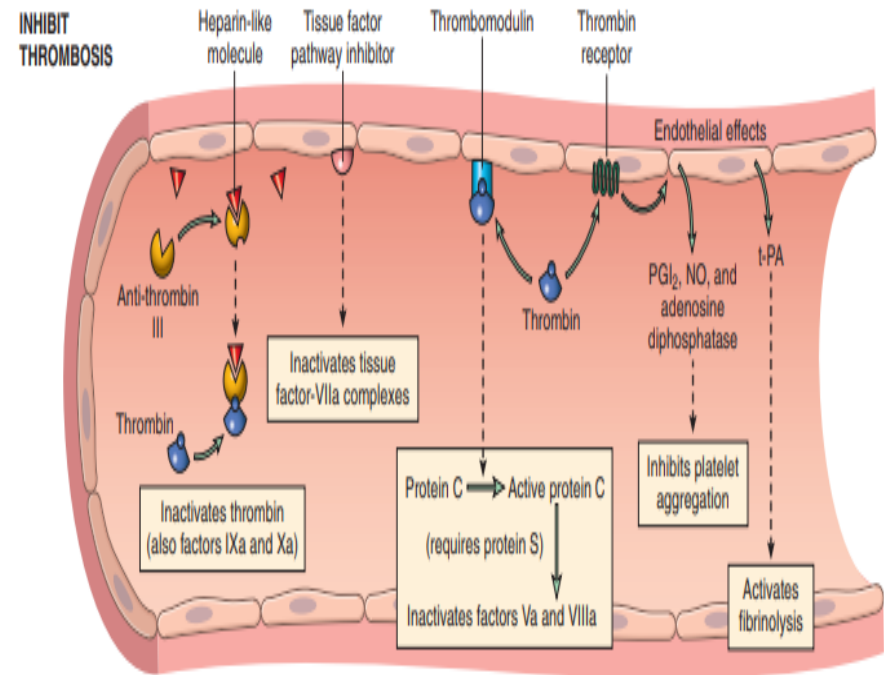


INHIBIT THROMBOSIS



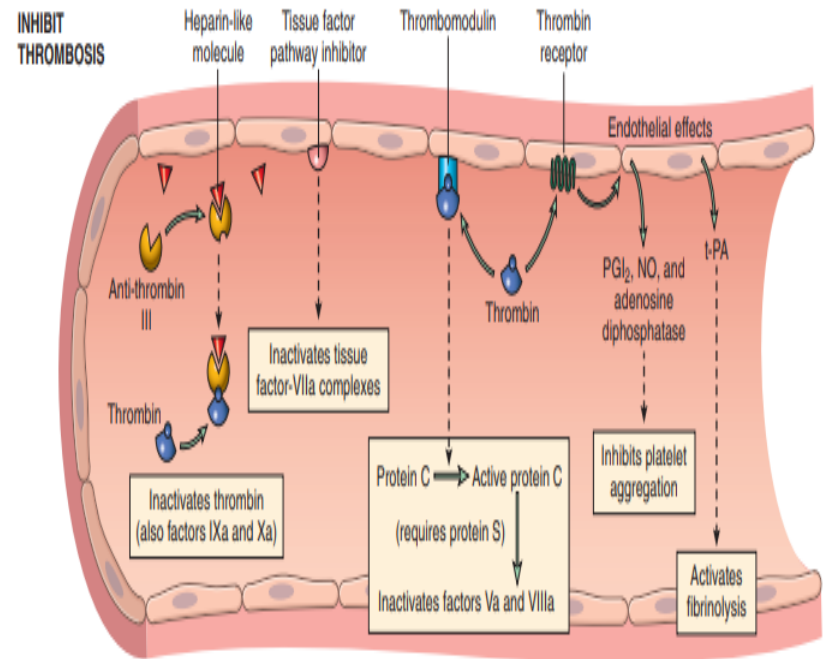
Heparin.

- By itself, it has little or no anticoagulant properties, but when it combines with antithrombin III, the effectiveness of antithrombin III for removing thrombin increases by a hundredfold to a thousandfold, and thus it acts as an anticoagulant.
- The complex of heparin and antithrombin III removes several other activated coagulation factors include factors XII, XI, X, and IX.
- largest quantities are formed by the mast cells located in the pericapillary connective tissue throughout the body.
- These cells continually secrete small quantities of heparin that diffuse into the circulatory system.
- The basophil cells of the blood release small quantities of heparin into the plasma



Antithrombin Action of Fibrin and Antithrombin III

- fibrin adsorbs thrombin and prevents the clotting reaction from spreading
- While a clot is forming, about 85 to 90 percent of the thrombin formed from the prothrombin becomes adsorbed to the fibrin fibers as they develop.
- This adsorption helps prevent the spread of thrombin into the remaining blood and, therefore, prevents excessive spread of the clot.
- The thrombin that does not adsorb to the fibrin fibers soon combines with antithrombin III, which further blocks the effect of the thrombin



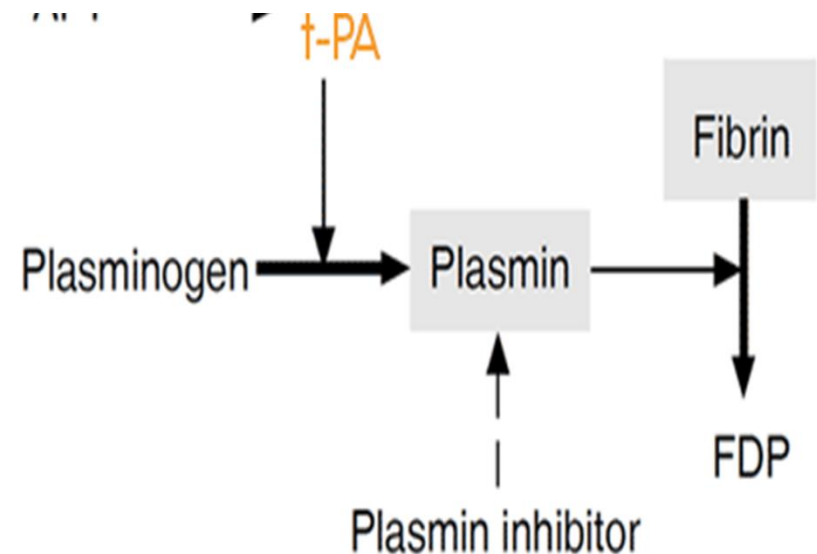
PREVENTION OF COAGULATION

TABLE 14.10 Factors That Inhibit Blood Clot Formation

Factor		Action	
Smooth lining of blood vessel	Prevents activation of intrinsic blood clotting mechanism	Antithrombin in plasma	Interferes with the action of thrombin
Prostacyclin	Inhibits adherence of platelets to blood vessel wall	Heparin from mast cells and basophils	Interferes with the formation of prothrombin activator
Fibrin threads	Adsorbs thrombin		

FIBRINOLYTIC MECHANISM

- Fibrinolysis is the system in which fibrin clot is dissolved as the vessel heals in order to restore normal blood flow
- Plasmin is a Proteolytic enzyme capable of degrading fibrin
- Plasmin is synthesized in the liver
- Plasmin normally present in plasma as plasminogen, is converted to Plasmin by the action of Plasminogen Activators, which is released by Damage Tissues
- Plasmin slowly dissolves fibrin clot with fibrin degradation products released in the plasma.



ANTI-PLASMIN

- Free Plasmin is released to the plasma, however, anti-plasmin is there immediately destroy any plasmin released from the clot.

FIBRINOLYTIC MECHANISM

FIBRIN DEGRADATION PRODUCTS

- In the process of fibrin degradation by plasmin within a clot, specific molecular fragments are produced called Fibrin Degradation Products (FDP)
- These degradation products are removed by the reticuloendothelial system and other organs.

Pathologic Effect of FDPs

- The FDPs are significant because of their haemostatic effects, which include;
- Anti-thrombin activity
- Interference with polymerization of fibrin monomer
- Interference with platelet activity

BLOOD CLOT

- The clot is composed of a meshwork of fibrin fibers running in all directions and entrapping blood cells, platelets, and plasma. The fibrin fibers also adhere to damaged surfaces of blood vessels; therefore, the blood clot becomes adherent to any vascular opening and thereby prevents further blood loss.
- **Clot Retraction and Expression of Serum.** Within a few minutes after a clot is formed, it begins to contract and usually expresses most of the fluid from the clot within 20 to 60 minutes. The fluid expressed is called **serum** because all its fibrinogen and most of the other clotting factors have been removed; in this way, serum differs from plasma.
- Contractile proteins, namely actin, myosin & thrombosthenin in the cytoplasm of platelets are responsible for clot retraction.

Formation of Plasmin

- Plasmin is formed from inactivated glycoprotein called plasminogen. Plasminogen is synthesized in liver and it is incorporated with other proteins in the blood clot.
- Plasminogen is converted into plasmin by tissue plasminogen activator (t-PA), lysosomal enzymes and thrombin. The t-PA and lysosomal enzymes are released from damaged tissues and damaged endothelium.
- Thrombin is derived from blood. The t-PA is always inhibited by a substance called t-PA inhibitor. It is also inhibited by factors V and VIII.
- Besides t-PA, there is another plasminogen activator called urokinase plasminogen activator (u-PA). It is
- derived from blood.

Sequence of Events Involved in the Activation of Plasminogen

1. During intravascular clotting, the endothelium of the blood vessel secretes a thrombin-binding protein, the thrombomodulin. It is secreted by the endothelium of all the blood vessels, except the minute vessels of brain.
2. Thrombomodulin combines with thrombin and forms a thrombomodulin-thrombin complex which activates protein C
4. Activated protein C inactivates factor V and VIII in the presence of a cofactor called protein S
5. Protein C also inactivates the t-PA inhibitor
6. Now, the t-PA becomes active. Activated t-PA and lysosomal enzymes activate plasminogen to form plasmin. Plasminogen is also activated by thrombin and u-PA

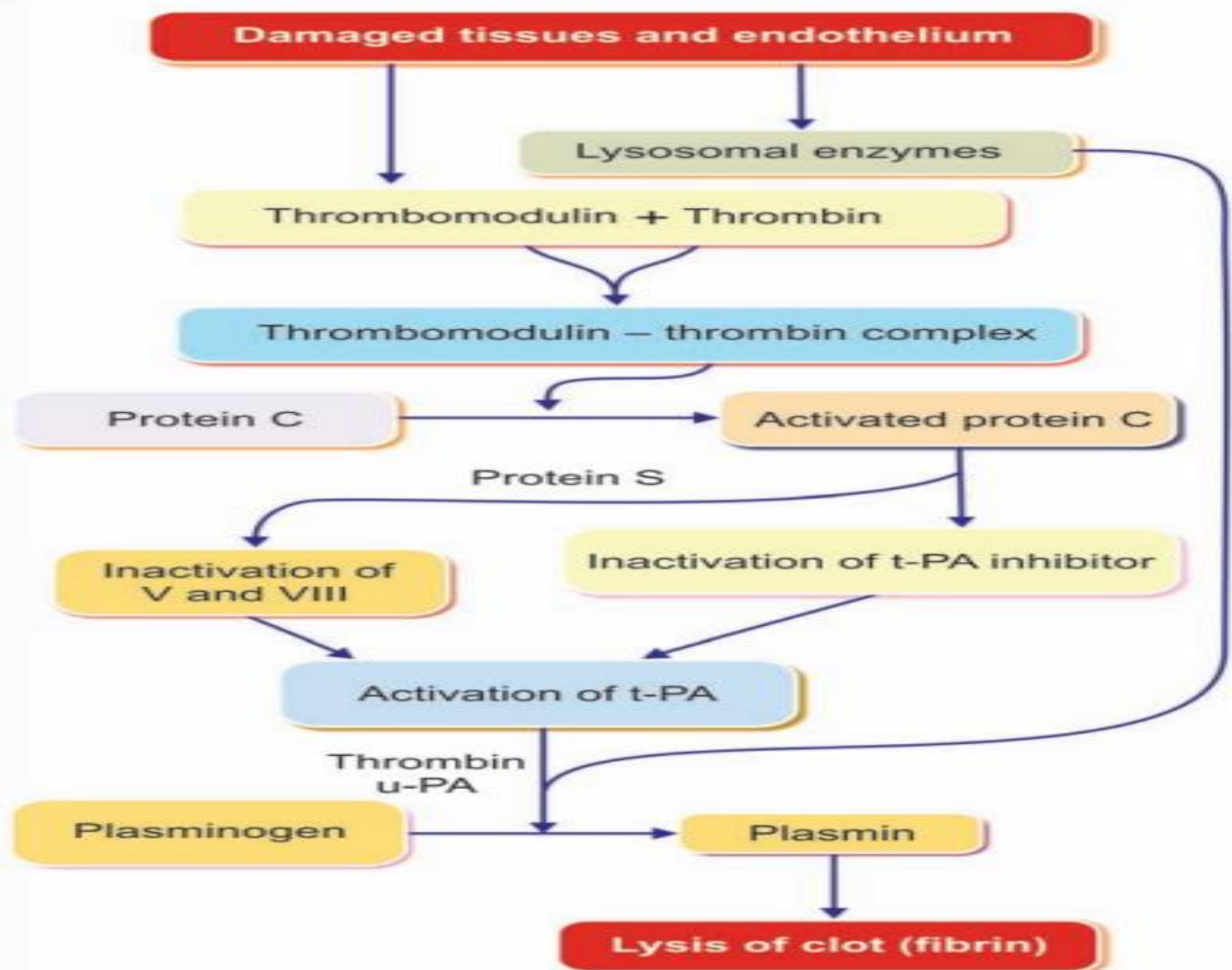


FIGURE 20.2: Fibrinolysis. t-PA = Tissue plasminogen activator, u-PA = Urokinase plasminogen activator.

TESTS FOR BLOOD CLOTTING

- Blood clotting tests are used to diagnose blood disorders.
- 1. Bleeding time
- 2. Clotting time
- 3. Prothrombin time
- 4. Partial prothrombin time
- 5. Thrombin time.
- **Bleeding time (BT)** is the time interval from oozing of
- blood after a cut or injury till arrest of bleeding. Usually,
- it is determined by Duke method using blotting paper or
- filter paper method. Its normal duration is 3 to 6 minutes.
- It is prolonged in purpura.

- **Clotting time (CT)** is the time interval from oozing of blood after a cut or injury till the formation of clot. It is usually determined by capillary tube method. Its normal duration is 3 to 8 minutes. It is prolonged in hemophilia
- **Prothrombin time (PT)** is the time taken by blood to clot after adding tissue thromboplastin to it. During this procedure, the time taken by blood to clot after adding tissue thromboplastin is determined. PT indicates the total quantity of prothrombin present in the blood.
- Normal duration of prothrombin time is 10 to 12 seconds. It is prolonged in deficiency of prothrombin and other factors like factors I, V, VII and X.

PARTIAL PROTHROMBIN TIME OR ACTIVATED PROTHROMBIN TIME

- Partial prothrombin time (PPT) is the time taken for the blood to clot after adding an activator such as phospholipid, along with calcium to it. It is also called activated partial prothrombin time (APTT). This test is useful in monitoring the patients taking anticoagulant drugs.
- Normal duration of partial prothrombin time is 30 to 45 seconds. It is prolonged in heparin or warfarin therapy (since heparin and warfarin inhibit clotting) and deficiency or inhibition of factors II, V, VIII, IX, X, XI and XII.

Practice questions

- To prevent blood loss after a tissue injury, blood vessels first
 - a. Form a platelet plug
 - b. Vasodilate
 - c. Initiate the coagulation cascade
 - d. Vasoconstrict *
 - e. Form a clot

2. Adhesion of platelets to exposed collagen of damaged tissues is due to the

- Thromboxane
- Serotonin
- ADP
- Von Willebrand factor*
- Thrombin

3.Extrinsic clotting mechanism is initiated by

- Hageman factor
- Fibrin-stabilizing factor
- Antihemophilic factor
- Tissue thromboplastin factor*
- Stuart factor

4. Intrinsic clotting mechanism is initiated by factor

- Tissue thromboplastin
- Fibrin stabilizing factor
- Prothrombin
- Fibrinogen
- Hageman factor

5. Vitamin K dependent coagulation factors are

- XII and XI
- VI and XIII
- VIII and III
- II, VII, IX, X
- I and IV